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Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code		
S - 0006 - UES		
Kurskod / Course code: DT081A	Provkod/ Test code: T101	Tentamensdatum / Examination date: 2024-10-29
Kursnamn / Course name: Datateknik AV, Signal- och bildbehandling		
Provnamn / Test name: Tentamen		

Skriv din anonymitetskod på varje inlämnat papper
Write your anonymous code on each sheet submitted

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	4,5		16			
2	X	2		17			
3	X	3		18			
4	X	0,5		19			
5	X	6		20			
6	X	4		21			
7	X	0,5		22			
8	X	3		23			
9	X	5		24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points		28,5	Betyg / Grade: D	Lärarsign./ Teachers sign			

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted	Inlämnad tentamen / Submitted exam	Leg kontroll / Control identification	Sign. tentamensvakt / Sign. invigilator
9	0	OK	CAD.

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Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2024-10-28 kl. 12:37:16

Example Solutions for DT081A – Signal and Image Processing Exam, 2024-10-29

1 (1 + 1 + 1 + 1 + 1 = 5 p)

a)
The process of converting a continuous-time signal into a discrete-time signal by taking periodic samples, $x[n] = x(nT_s)$ where T_s is the sampling interval.

b)
The process of mapping a continuous range of values to a finite set of discrete levels, often in analog-to-digital conversion.

c)
The output of a system when an impulse signal is applied at its input. It characterizes the system's behavior.

d)
The steady-state response of a system to sinusoidal inputs at varying frequencies.

e)
A technique used to reduce spectral leakage in signal analysis by applying a window function to a signal before performing a Fourier transform.

2 (3 + 2 = 5 p)

a)
$y[n] = s[n] * h[n]$

$y[0] = (3)(-0.5) = -1.5$
$y[1] = (7)(-0.5) + (3)(0) = -3.5$
$y[2] = (10)(-0.5) + (7)(0) + (3)(0.5) = -5 + 1.5 = -3.5$
$y[3] = (3)(-0.5) + (10)(0) + (7)(0.5) = -1.5 + 3.5 = 2$
...

$y[n] = \{-1.5, -3.5, -3.5, 2, 7.5, 1.5, -2.5\}$

b) This filter approximates a differentiation operation, emphasizing changes in the signal, which is characteristic of a high-pass filter.

3 (1 + 2 = 3 p)

a) The system diagram shows both a feed-forward and feedback branch. The latter implies that the system will produce an infinite impulse response IIR.

b)
$y(n) = x(n) + 2x(n-1) + x(n-3) - 3y(n-1) + 2y(n-3)$

4 (2 + 3 = 5 p)

a)
Extend spectrum plot to at least $-2\pi < \Omega < 2\pi$, where the continous spectrum is mirrored on the y-axis and replicated over the whole Ω axis, with a periodicity of 2π.

b)
$f_s \geq 2f_{max}$
The lowest sampling frequency depends on the signal bandwidth. The figure shows that $f_{max} = \frac{3}{4}\pi\frac{f_s}{2\pi} = \frac{3}{8}f_s = \frac{150}{8} = 18.75Hz$.
Hence $f_s \geq 37.5Hz$.

5 (2 + 4 = 6 p)

a) Amplitude and phase relationship:
$|Y(f)| = |H(f)||X(f)|$
$\angle Y(f) = \angle H(f) + \angle X(f)$

b) Sketch spectra

$$|Y(f)| = \begin{cases} 8f & \text{for } 0 \leq f \leq 10 \\ 80 - 4(f-10) & \text{for } 10 < f \leq 30 \\ 0 & \text{for } f < 0 \text{ or } f > 30 \end{cases}$$

and

$arg(Y(f)) = 0 \forall f$

6 (2 + 2 = 4 p)

a) Window iii) has the narrowest main lobe, which provides the best frequency separation between closely spaced frequency components allowing for better distinction.
b) Window iv) has zero outside of main lobe, which effectively eliminates spectral leakage in the frequency domain ensuring that frequency components are not contaminated by artifacts from distant frequencies.

7 (3 p)

At low frequencies the system has an amplification of 1, whereas higher frequencies are amplified more. Combined this results in high frequency amplification (not to be confused with a high pass filter), which if applied to an image would enhance edges.

8 (1 + 1 + 1 + 1 + 1 = 5 p)

- $1 \rightarrow ii$
  Only the center element is used, and reduce the input with 0.5.
- $2 \rightarrow iv$
  A vertical filter emphasized vertical edges only.
- $3 \rightarrow v$
  Edge detecting kernel with elements summing to 0, only shows edges.
- $4 \rightarrow iii$
  Avering kernel smooths the image as a low-pass filter.
- $5 \rightarrow i$
  Edge detecting kernel summed with a unit kernel translates kernel to one with center element = 5, with elements summing to 1.

9 (6 p)

$x(i,j)$ has several non-zero values, that each contribute to the $y(i,j)$ through convolution with $h(i,j)$. From examining the isolated scaled impulse responses we can determine that $h(i,j)$ has a cross-shaped pattern. By subtracting one scaled impulse reponse from another we can identify the impulse response as.

$$h(i,j) = \begin{bmatrix} 0 & 1/8 & 0 \\ 1/8 & 1/2 & 1/8 \\ 0 & 1/8 & 0 \end{bmatrix}$$

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Anonymitetskod/ Anonymus code: S-0006-UES

Sidnr / Page no: 1

Lärarens anteckning / Teachers note:

1.

a) Sampling is the process of taking values of a continious signal at a certain rate. Used to turn a continous signal to discrete. 1

b) Quantization is taking a signal and reducing the amount of bits that is used to represent the signal 0,5

c) Impulse response is the response a system gives when an impulse is given as imput. The impulse response defines the system. 1

d) Frequency response is the response of a system that defines how much a frequency component is adjusted in magnitude and phase-shift. 1

e) Windowing is a method used to limit and "capture" a certain part of a signal. By applying a window function to a signal, the signal part which lays inside the boundaries of the window stay the same but part outside is "filtered away to 0. ↖ idealy 1

4,5

Skriv ej på baksidan! Do not write on the other side!

Lärarens anteckning / Teachers note:

2 a) $s(n) = \{3, 7, 10, 3, -5\}$

$h(n) = \{-0,5;\ 0;\ 0,5\}$

$y(n) = \sum_{k=-\infty}^{\infty} s(k) \cdot h(n-k)$

$y(0) = s(0) \cdot h(0-0) = 3 \cdot (-0,5) = \underline{-1,5}$

$y(1) = s(0) \cdot \underbrace{h(1-0)}_{=0} + s(1) \cdot h(1-1) = 7 \cdot (-0,5) = \underline{-3,5}$

$y(2) = s(0) \cdot h(2-0) + s(1) \cdot \underbrace{h(2-1)}_{=0} + s(2) \cdot h(2-2)$

$= 1,5 + (-5) = \underline{-3,5}$

$y(3) = \cancel{s(0) \cdot h(3-0)} + s(1) \cdot h(3-1) + s(2) \cdot \underbrace{h(3-2)}_{=0}$

$+ s(3) \cdot h(3-3) = 3,5 + (-1,5) = \underline{2}$

assume $h(n) = 0$, $0 > n \geq 3$

$y(4) = \cancel{s(0) \cdot h(4-0)} + \cancel{s(1) \cdot h(4-1)} + s(2) \cdot h(4-2)$

$+ s(3) \cdot \underbrace{h(4-3)}_{=0} + s(4) \cdot h(4-4)$

$= 5 + 2,5 = \underline{7,5}$

$y(n) = \{-1,5; -3,5; -3,5; 2; 7,5\}$

2

b) It almost inverts the signal ?

1

0

/2

Lärarens anteckning / Teachers note:

3 a) This system is a LTI system that is causal but it has an infinite impulse response (IIR) meaning it is unstable ?.

1

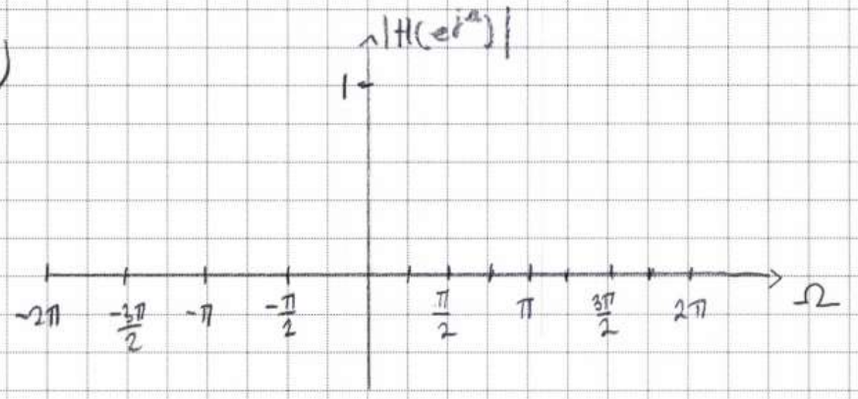
b) $y(n) = x(n) + 2x(n-1) + x(n-3) - 3y(n-1) + 2y(n-3)$

2

/3

Lärarens anteckning / Teachers note:

4. a)



0

b) $f_s = 2 \cdot f_{max}$ is the lowest (Nyquist-sharma)

0,5

0,5

Lärarens anteckning / Teachers note:

5 a) $|Y(f)| = |X(f)| \cdot |H(f)|$

$\arg Y(f) = \arg X(f) + \arg H(f)$

2

Interval f

$0 \le f < 10$

$|X(f)| = 0,4 \cdot f$ $\quad \arg X(f) = \frac{\pi}{40} \cdot f$

$|H(f)| = 20$ $\quad \arg H(f) = -\frac{\pi}{40} \cdot f$

$|Y(f)| = 0,4 \cdot f \cdot 20$ $\quad \arg Y(f) = 0$

$10 \le f < 20$

$|X(f)| = 4$ $\quad \arg X(f) = \frac{\pi}{4}$

$|H(f)| = 30 - f$ $\quad \arg H(f) = -\frac{\pi}{4}$

$|Y(f)| = 4 \cdot (30 - f)$ $\quad \arg Y(f) = 0$

$20 \le f \le 30$

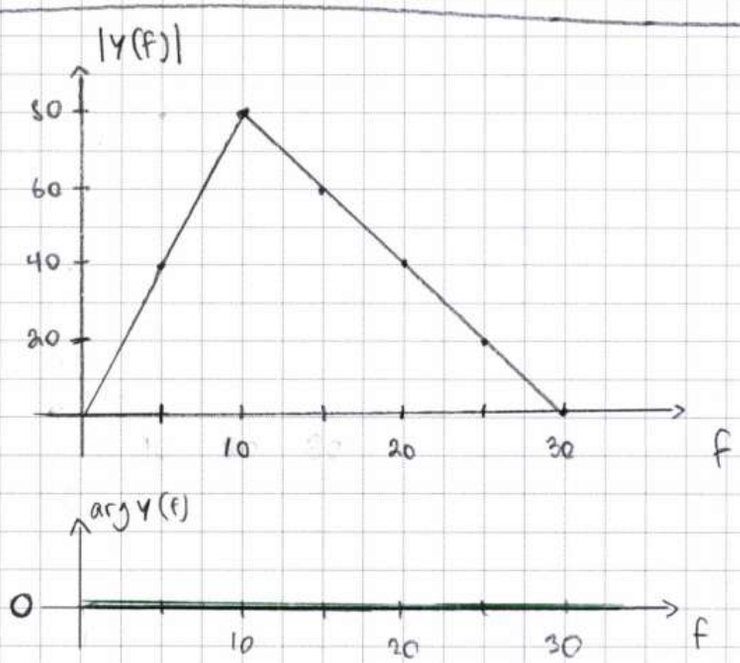
$|X(f)| = -0,4 \cdot f + 12$ $\quad \arg X(f) = \frac{\pi}{40} \cdot f - \frac{\pi}{4}$

$|H(f)| = 10$ $\quad \arg H(f) = -\frac{\pi}{40} \cdot f + \frac{\pi}{4}$

$|Y(f)| = (-0,4 \cdot f + 12) \cdot 10$ $\quad \arg Y(f) = 0$

$F > 30 \quad X = ?$

b.)



4

1/6

Lärarens anteckning / Teachers note:

6 a) The window that achieves the best frequency separation is the window ~~that~~ produces the narrowest main lob. This would be iii). The wider the lobe the more chance of leakage into another freq component 2

b) The window function that minimizes spectral leakage is the window function that introduces the least amount of oscillation (side lobes). This would be iv) 2

/4

Lärarens anteckning / Teachers note:

7. This system could be compared to the following kernel:

6,3	5	6,3
5	1	5
6,3	5	6,3

This is a system that increases the magnitude of low intensity elements by a lot and high intensity elements by a little if they're close. This would smoothen out the input. ^ creating a lower freq output.

0,5

Lärarens anteckning / Teachers note:

8. 1)- ii) Kernel 1 reduces the pixel intensity for every pixel by half making the image darker. 1

2)- i) Kernel 2 sharpens the image 0

3)- v) kernel 3 is an edge-detection kernel that enhances the edges of the image by increasing the intensity of already high intensity pixels "more" than low intensity pixels and then flipping them to negative values. 1

4)- iii) kernel 4 introduces a blur to the image by averaging out the values of the pixels. 1

5)- iv) kernel 5 is very similar to kernel 4 and does the same thing but with a different increase of pixel intensity. Unsure if kernel 5 matches to image iv) or v) 0

/3

Lärarens anteckning / Teachers note:

9.

$h(i,j)$

	0	1	2
0	0	0,125	0
1	0,125	0,5	0,125
2	0	0,125	0

Motivation: A lot of information about the kernel $h(i,j)$ above can be gathered by comparing the top left 3×3 "sub-matrix" of the input matrix $x(i,j)$ and the output matrix $y(i,j)$. We see that the center 112 becomes 56 which is half, we see that values above, under, to the left and to the right of the center goes from 0 to $\frac{112}{8} = 14$. $\frac{1}{8} = 0,125$ We also see that the top left and right corner remains 0. By applying the kernel we have so far on $x(3,0)$, $x(3,2)$, $x(2,3)$ we can also confirm that the bottom left and right of $h(i,j)$ also is 0. By applying $h(i,j)$ on $x(i,j)$ we get $y(i,j)$

5